TRAVERSE ADJUSTMENT REPORT

SurveyX Digital
Date: 2026-10-23

PROJECT	**Al Khor Expressway**	LOCATION	**Al Khor, Qatar**
SURVEYOR	**Asjad Mehmood**	CHECKED BY	**Senior Surveyor**
INSTRUMENT	**Leica TS16 2"**	DATE	**2026-10-23**

1. Traverse details

Traverse type	**Closed (checked on a known point)**
Observations	**Field book: angles turned clockwise from the backsight, and distances**
Adjustment method	**Bowditch (Compass rule)**
Coordinate system	**Qatar National Grid (QND95), WGS84 latitude and longitude**
Scale factor	**0.99999941 = point 1.00000019 x height 0.99999922 (5.0 m)**
Station point	**CP-01 E 228450.312 N 441203.845**
Backsight point	**CP-02 E 228912.640 N 441655.118**
Reference bearing	**CP-01 -> CP-02 045°41'36" (646.061 m)**
Closing point	**CP-01 E 228450.312 N 441203.845**
Closing angle	**23.1021 at CP-01, onto CP-02**
Traverse legs	**6 (total length 1810.361 m)**

2. Closure check

ACCEPTED - within every tolerance

Misclosure dE = 10.0 mm dN = -10.4 mm Linear = 14.4 mm
Precision achieved 1 : 125,384 Required 1 : 20,000 Total length 1810.361 m
Angular misclosure +2.0" on 7 angles Allowed +/-13.2" (5" x sqrt n) Correction -0.3" per angle
Level misclosure -7.0 mm Allowed +/-10.8 mm
Misclosure = where the observations arrive - known closing coordinates. Corrections are the opposite sign.

3. Field book (as observed)

#	Kind	Angle	Distance (m)	Ht diff (m)	Point	Remark
1	Point	81.3127	316.630	0.646	TS-01	Nail in kerb
2	Side shot	312.4500	42.315	-0.212	MH-01	Manhole cover
3	Point	124.3129	308.240	0.584	TS-02	
4	Point	125.5354	308.364	0.566	TS-03	
5	Side shot	201.1530	27.904	0.118	LP-01	Lamp post
6	Point	104.5917	293.903	-0.802	TS-04	
7	Point	129.2554	294.528	-0.478	TS-05	
C	Closing leg	130.2740	288.697	-0.523	CP-01	Closing leg

4. Computed coordinates (angles balanced, before the linear adjustment)

Point	Easting (m)	Northing (m)	Remark
CP-01	228450.312	441203.845	Station (known)
TS-01	228702.459	441012.334	Nail in kerb

Point	Easting (m)	Northing (m)	Remark
TS-02	228995.181	441108.905	
TS-03	229088.627	441402.769	
TS-04	228841.106	441561.237	
TS-05	228560.899	441470.515	
CP-01	228450.322	441203.835	Closing point as observed

5. Output coordinates (adjusted)

Point	Corr. dE (mm)	Corr. dN (mm)	Adjusted Easting (m)	Adjusted Northing (m)	Level (m)
CP-01	0.0	0.0	**228450.312**	**441203.845**	4.215
TS-01	-1.8	1.8	**228702.457**	**441012.336**	4.862
TS-02	-3.5	3.6	**228995.177**	**441108.908**	5.447
TS-03	-5.2	5.4	**229088.622**	**441402.774**	6.015
TS-04	-6.8	7.0	**228841.099**	**441561.244**	5.214
TS-05	-8.4	8.7	**228560.890**	**441470.524**	4.737
CP-01	-10.0	10.4	**228450.312**	**441203.845**	4.215

6. Bearings and distances (from output coordinates)

From	To	Bearing (WCB)	Distance (m)	Observed (m)
CP-01	TS-01	127°13'02"	316.627	316.630
TS-01	TS-02	071°44'30"	308.239	308.240
TS-02	TS-03	017°38'23"	308.365	308.364
TS-03	TS-04	302°37'42"	293.905	293.903
TS-04	TS-05	252°03'37"	294.529	294.528
TS-05	CP-01	202°31'17"	288.696	288.697

7. Angles (clockwise from the backsight)

At	Backsight	Foresight	Observed	Correction	Adjusted
CP-01	CP-02	TS-01	081°31'27.0"	-0.3"	081°31'26.7"
TS-01	CP-01	TS-02	124°31'29.0"	-0.3"	124°31'28.7"
TS-02	TS-01	TS-03	125°53'54.0"	-0.3"	125°53'53.7"
TS-03	TS-02	TS-04	104°59'17.0"	-0.3"	104°59'16.7"
TS-04	TS-03	TS-05	129°25'54.0"	-0.3"	129°25'53.7"
TS-05	TS-04	CP-01	130°27'40.0"	-0.3"	130°27'39.7"
CP-01	TS-05	CP-02	023°10'21.0"	-0.3"	023°10'20.7"

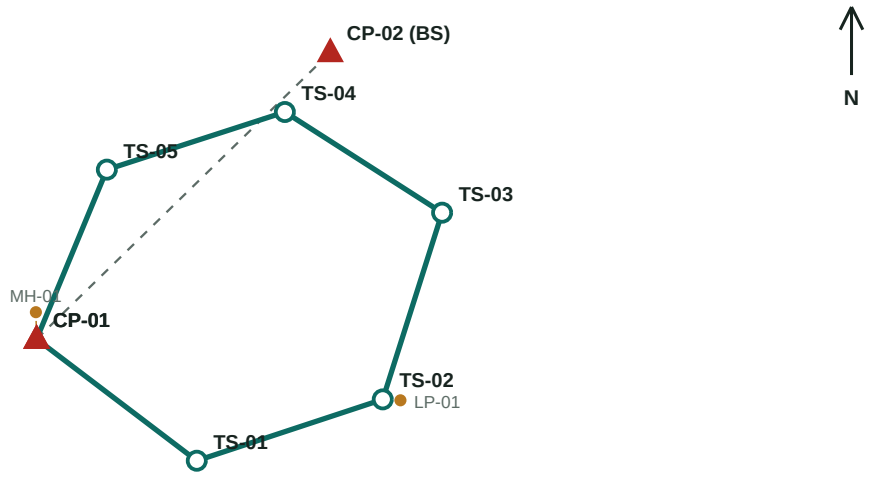
8. Side shots (from the adjusted stations)

Point	From	Bearing	Distance (m)	Easting (m)	Northing (m)	Level (m)	Remark
MH-01	CP-01	358°26'36"	42.315	**228449.162**	**441246.144**	4.003	Manhole cover
LP-01	TS-02	093°00'00"	27.904	**229023.043**	**441107.448**	5.565	Lamp post

9. Latitude and longitude (WGS84, from the Qatar National Grid)

Point	Easting (m)	Northing (m)	Latitude	Longitude
CP-01	228450.312	441203.845	25.72512657	51.49940387
TS-01	228702.457	441012.336	25.72339300	51.50191242
TS-02	228995.177	441108.908	25.72425898	51.50483149
TS-03	229088.622	441402.774	25.72690975	51.50576909
TS-04	228841.099	441561.244	25.72834507	51.50330587
TS-05	228560.890	441470.524	25.72753164	51.50051151
MH-01	228449.162	441246.144	25.72550841	51.49939331
LP-01	229023.043	441107.448	25.72424525	51.50510914

10. Sketch



Sketch, not to a printed scale. Red triangles: known points. Circles: traverse points (adjusted).

SURVEYED BY	CHECKED BY	APPROVED BY
Asjad Mehmood	**Senior Surveyor**	
Signature / Date	Signature / Date	Signature / Date